

FINAL EXAMINATION

Section A

Q1. (1 marks)

Given the following self-join query, identify which subquery type is used in the WHERE clause and briefly explain its purpose.

The subquery used in the WHERE clause is an example of a `Predicate Subquery`. Its purpose is to test if a row exists in another table (sp) with the same part number (pno), different supplier numbers (sno), and non-null quantities.

Q2. (1 marks)

Given the SQL query:

```
SELECT * FROM sp a WHERE qty > ALL (SELECT qty FROM sp b WHERE a.pno = b.pno AND a.sno <> b.sno AND qty IS NOT NULL)
```

How does this query find the supplier with the highest quantity of each part, ignoring null quantities?

Q3. (1 marks)

Given the table 'sp' containing part number (pno), supplier number (sno), quantity, description, and color. Write a SQL query that lists the suppliers who have a higher quantity for each part than any other supplier (ignoring null quantities).

Q4. (1 marks)

Which of the following subquery types allows several types of tests and can use the full set of comparison operators, while correlated subqueries must enclose their queries in parentheses? Also, which type requires retrieving only one column in its select list?

Q5. (1 marks)

```
{
  "requires_figure": false,
  "fingerprint": "subqueries-self_join",
  "blocks": [
    { "type": "text", "text": "Given the context about subqueries and self joins, consider the following SQL statement: \n\nSELECT * FROM sp a WHERE qty > ALL (SELECT AVG(qty) FROM sp b WHERE a.pno = b.pno AND a.sno <> b.sno)\n\nWhat can be inferred from this query about the part numbers (pno) that have an average quantity greater than their individual quantities in the 'sp' table?"
    }
  ]
}
```

Q6. (1 marks)

Given two tables, p and sp, consider the following SQL queries. Identify the type of subquery (correlated or predicate) in each case and explain why.

(1) `SELECT a.pno FROM sp a WHERE a.pno = (SELECT b.pno FROM sp b WHERE a.sno <> b.sno);`

(2) `SELECT * FROM sp a WHERE qty > ALL (SELECT qty FROM sp b WHERE a.pno = b.pno AND a.sno <> b.sno AND qty IS NOT NULL);`

Q7. (1 marks)

What SQL query would you use to list the part number (pno) and the supplier number (sno) of suppliers who have the highest quantity for each part, ignoring null quantities?

And how would you write this query using a self join?

Q8. (1 marks)

Which of the following subquery types tests if a subquery retrieves at least one row?

Q9. (1 marks)

Which of the following is NOT true about subqueries in SQL?

1. IN Subquery can test whether a scalar value matches any subquery result row.
2. Predicate subquery must retrieve one column in their select list and can be used in the WHERE clause.
3. Self join is a type of predicate subquery that tests for the highest quantity of each part ignoring null quantities.

Q10. (1 marks)

Which of the following keywords indicates a subquery that uses correlation names to reference columns in both the outer and inner query? (Multiple options can be chosen.)

Q11. (1 marks)

Given the context about various types of subqueries, can you identify and explain the general format for IN Subqueries and EXISTS Subqueries in SQL?

IN Subquery checks if a scalar value matches any single query column value within a subquery result row. Its general format is as follows: ``value [NOT] IN (query)``. Can you provide an example of how it's used in SQL?

EXISTS Subquery tests whether a subquery retrieves at least one row, meaning whether the subquery has any results. Its general format is: ``EXISTS (query)``. Can you provide an example of how it's used in SQL?

Q12. (1 marks)

What type of subquery allows several types of tests and can use the full set of comparison operators? And what is its general format?

Q13. (1 marks)

Given the context of self joins and correlated subqueries, consider the following query: ``SELECT * FROM sp a WHERE qty >ALL (SELECT qty FROM sp b WHERE a.pno = b.pno AND a.sno <> b.sno AND qty IS NOT NULL)``. What does this query do?

Q14. (1 marks)

```
{  
  "requires_figure": false,  
}
```

```

"fingerprint": "correlated_subqueries-self_join",
"blocks": [
  {"type": "text", "text": "Given the following self-join query: `SELECT * FROM sp a WHERE qty >ALL (SELECT qty FROM sp b WHERE a.pno = b.pno AND a.sno <> b.sno AND qty IS NOT NULL)` identify the role of subquery in this query."},
  {
    "type": "text", "text":
      "a) Returning all rows that have higher quantity than any other row with the same part number (ignoring null quantities).\n" +
      "b) Checking if there exist rows in `sp` with the same part number, different supplier numbers, and non-null quantities.\n" +
      "c) Joining rows in `sp` based on part number while ignoring rows with null quantities.\n" +
      "d) Returning all rows that have the highest quantity for each part number."
  }
]
}

```

Q15. (1 marks)

Given the following database schema and sample data, determine if there exists any supplier (pno) who provides a specific color ('Red') for the part with pno 'P2'. Use EXISTS subquery to solve this problem.

The database schema and sample data are as follows:

sp(sno, pno, qty, descr, color)

```

[{"sno": "S1", "pno": "P1", "qty": NULL, "descr": "Widget", "color": "Blue"}, {"sno": "S2", "pno": "P1", "qty": 200, "descr": "Widget", "color": "Blue"}, {"sno": "S3", "pno": "P1", "qty": 1000, "descr": "Widget", "color": "Blue"}, {"sno": "S3", "pno": "P2", "qty": 200, "descr": "Widget", "color": "Red"}]

```

Write the SQL query using EXISTS subquery to find the answer.

The correct SQL query would be:

```

SELECT EXISTS (SELECT 1 FROM sp WHERE pno = 'P2' AND color = 'Red') AS result

```

Q16. (1 marks)

Given the context of self-joins and correlated subqueries, consider the following SQL query:

```

SELECT * FROM sp a WHERE qty >ALL (SELECT qty FROM sp b WHERE a.pno = b.pno AND a.sno <> b.sno AND qty IS NOT NULL)

```

What does this query do? It finds the suppliers with quantities for each part that are greater than all other quantities of the same part, ignoring null quantities.

Q17. (1 marks)

Consider the following 'sp' table that represents supply parts. Each row contains the supplier number (sno), part number (pno), quantity, description, and color.

Write a SQL query to find the supplier with the highest quantity of each part, ignoring null quantities, using an EXISTS subquery in the WHERE clause.

Q18. (1 marks)

Consider the following SQL query: 'SELECT * FROM sp a WHERE EXISTS (SELECT * FROM sp b WHERE a.pno = b.pno AND a.sno <> b.sno AND qty IS NOT NULL)'.

What does this query do, and how does it differ from other subqueries?

The given SQL query is an example of an EXISTS Subquery. Unlike other subqueries, the EXISTS Subquery tests whether a subquery retrieves at least one row, rather than returning a value or set of values.

In this specific case, the EXISTS Subquery is used to filter rows in the 'sp' table where there exists another row with the same part number (pno), different student number (sno), and non-null quantity. This query helps find suppliers who have supplied at least one non-null quantity of a specific part.

This EXISTS Subquery can be particularly useful when you want to check for the existence of related data without needing to know its exact values.

Q19. (1 marks)

```
{
  "requires_figure": false,
  "fingerprint": "predicate_subqueries-exists",
  "blocks": [
    {
      "type": "text",
      "text": "Which SQL query uses the EXISTS subquery to check if there are any suppliers for part P1 who have provided more than 500 units of that part?",
    },
    {
      "type": "figure",
      "figure_id": "fig_0010"
    },
    {
      "type": "text",
      "text": "Given the context, the correct SQL query would be:"
    }
  ]
}
```

```
SELECT *
FROM sp
WHERE EXISTS (SELECT 1 FROM sp b WHERE a.pno = b.pno AND b.pno = 'P1'
AND qty > 500)}
```

Q20. (1 marks)

Given the context about subqueries and self-joins, consider the following scenario: A table 'parts' contains information about different parts with columns pno (part number), sno (supplier number), qty (quantity), descr (description), and color. Another table 'orders' contains information about orders with columns ono (order number), pno, qty_ordered, and total_cost. Write a SQL query to find the parts for which the quantity ordered is greater than the quantity in stock.

Section B

Q21. (2 marks)

Given that total participation of an entity set in a relationship set is represented by a thick line and partial participation by a single line, how can you represent the partial participation of dependents in policy?

Q22. (2 marks)

In the given context, how can we identify total participation of an entity set in a relationship set and partial participation?

Q23. (2 marks)

Given the context of ER diagrams, explain how 'Aggregation' and 'Partial Participation' differ in terms of representation.

Q24. (2 marks)

What is the difference between 'Aggregation' and 'Partial Participation' in Entity-Relationship Diagrams?

Q25. (2 marks)

Describe what a 'weak Entity Set' is in ER diagrams.

Q26. (2 marks)

What is the difference between aggregation and weak entities in ER modeling?

Section C

Q27. (3 marks)

Given the function examples provided, can you write an example of a function in SQL that takes two arguments and returns their sum?

Q28. (3 marks)

Write the equivalent expression using logical operators for the given condition: `sno = 'S1' OR city = 'Rome' AND has_car = True`.

Q29. (3 marks)

Which of the following expressions has a higher precedence: `'a AND b OR c'` or `'a OR (b AND c)'`?

Q30. (3 marks)

In the given logical expression ``a OR b AND c``, if we want to change its precedence to match that of the parentheses, what should be the equivalent expression using parentheses?

Q31. (3 marks)

Given the logical operator precedence rules, which of the following expressions is equivalent to `a OR b AND c`?

Q32. (3 marks)

```

{
  "requires_figure": false,
  "fingerprint": "logical_expression_application",
  "blocks": [
    {
      "type": "text",
      "text": "Given the logical expressions: 'a AND b OR c NOT d' and 'e OR (f AND g)' with the following variables defined: a=True, b=False, c=True, d=False, e=False, f=True, g=True. Evaluate both expressions."
    },
    {
      "type": "text",
      "text": "a) What is the result of 'a AND b OR c NOT d'? \n" +
        "b) What is the result of 'e OR (f AND g)'?"
    }
  ]
}

```

Section D

Q33. (4 marks)

Given the context about CASE expressions, write a SQL query that uses a CASE expression and the LIKE operator to select names from the 's' table where the city is either 'Rome' or 'Paris', but only if the name contains the letter 'a'. Use the '%' wildcard character to achieve this.

Q34. (4 marks)

In a SQL query, given the context of expression operators and logical predicates, can you write a CASE clause that filters students' grades based on their exam results? The CASE clause should return the grade 'A' for students who scored 90 or above, 'B' for scores between 80 and 89, and 'C' or null for all other scores. Hint: Use the '=' operator and wild-card characters if necessary.

Q35. (4 marks)

Given the two forms of CASE statements and their functioning, write a SQL query using the LIKE operator to find all names from the 's' table where the name contains the substring 'a' (using the wildcard character '%').

Section E

Q36. (5 marks)

Given the OR operator logic and truth table provided, construct a question that requires understanding its operation by providing two different scenarios where the output of the OR operator is true, false, and unknown.

Q37. (5 marks)

Given the OR operator's functionality, construct an SQL query using the OR operator to retrieve all records where either 'pno' is equal to 'P2' or 'qty' is greater than 200 from a table named 'Department of IT, SVECW'.

Section F

Q38. (8 marks)

Given the context about functions and operators, write a SQL query that uses the OR operator (OR) to select rows from a table 'sp' where either the sno is 'S3' or the quantity is less than 500.